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**Psychopathy: Current
Knowledge and Future
Directions**

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Abstract

Research on psychopathy has progressed considerably in recent years against the backdrop of important advances in the broader field of clinical psychological science. My major aim in this review is to encourage integration of investigative work on dispositional, biobehavioral, and developmental aspects of psychopathy with counterpart work on general psychopathology. Using the triarchic model of psychopathy as a frame of reference, I offer perspective on long-standing debates pertaining to the conceptualization and assessment of psychopathy, discuss how dispositional facets of psychopathy relate to subdimensions of internalizing and externalizing psychopathology, and summarize findings from contemporary biobehavioral and developmental research on psychopathy. I conclude by describing a systematic strategy for coordinating biobehavioral-developmental research on psychopathy that can enable it to be informed by, and help inform, ongoing research on mental health problems more broadly.

Contents

INTRODUCTORY OVERVIEW	388
PERSISTING DEBATES IN THE STUDY OF PSYCHOPATHY: CURRENT KNOWLEDGE AND UNRESOLVED ISSUES	389
Issues of Definition and Measurement	389
Heterogeneity of Psychopathy	397
Psychopathy and Other Mental Disorders	398
UNDERSTANDING PSYCHOPATHY IN BIOBEHAVIORAL-DEVELOPMENTAL TERMS	401
CONCLUSIONS AND RECOMMENDATIONS FOR FURTHER ONGOING RESEARCH	403
Conceptualization and Assessment	403
Developmental Research	403
Neurobiological Research	404

INTRODUCTORY OVERVIEW

I began studying the phenomenon of psychopathic personality (psychopathy) as a clinical PhD student at the University of British Columbia in the mid-1980s, and I have continued researching this fascinating topic up to the present. My first investigation in this area, conducted for my dissertation, was a study examining physiological detection of deception (lie detector testing) in incarcerated offenders diagnosed as psychopathic (Patrick & Iacono 1989). Following from this initial study, a major focus of my investigative work over the past 35+ years has been the use of brain and other physiological measures to understand psychopathy.

My purpose in this review is to summarize what we have learned about this topic over the years since my initial study, with an emphasis on research conducted over the past two decades, and to suggest profitable directions for future investigation. The **Supplemental Appendix** to this article begins with a summary of some notable developments in the field of clinical science more broadly—namely, (a) the formulation of an integrative dimensional model of psychopathology that encompasses both episodic clinical conditions and personality disorders (PDs), the Hierarchical Taxonomy of Psychopathology (HiTOP; Kotov et al. 2017); (b) funding agency initiatives calling for increased use of neurobiological measures to clarify the role of brain processes in mental health problems and to improve prevention and treatment methods; (c) ascendance of a developmental perspective, which considers the interplay of genetic and environmental influences across time and views trait dispositions, and their relations with psychopathology, as continually evolving rather than static; and (d) integration of measures across large-scale etiologically informative studies, through use of data harmonization, to advance understanding of mental health problems.

Against the backdrop of these broader developments in the field, I consider persisting debates in the psychopathy area in relation to an integrative conceptualization of psychopathy, the triarchic model (Patrick et al. 2009), and review research that shows how subdimensions of psychopathy described by this model relate to dimensions of general psychopathology. Next, I discuss contemporary research aimed at understanding psychopathy in biobehavioral and developmental terms. I conclude by describing a systematic strategy for coordinating biobehavioral-developmental research on psychopathy that can allow it to be informed by, and in turn help to inform, ongoing research on mental health problems more broadly.

Supplemental Material >

PERSISTING DEBATES IN THE STUDY OF PSYCHOPATHY: CURRENT KNOWLEDGE AND UNRESOLVED ISSUES

A conceptual framework that is helpful for assimilating different theoretical perspectives and integrating findings across studies using different measures of psychopathy is the triarchic model (Patrick et al. 2009, Patrick & Drislane 2015b, Sellbom 2018). This model characterizes psychopathy in terms of three trait constructs that correspond to distinct symptom features of psychopathy but relate more clearly to biobehavioral systems and processes. These are (a) boldness, which encompasses social dominance, venturesomeness, and emotional resilience and connects with the biobehavioral process of threat sensitivity; (b) meanness, which entails low empathy, callousness, and aggressive manipulation of others and relates to biobehavioral systems for affiliation (social connectedness and caring); and (c) disinhibition, which involves boredom proneness, lack of restraint, irritability, and irresponsibility and relates to the biobehavioral process of inhibitory control. **Supplemental Table 1** lists various scale measures that have been developed to operationalize the trait constructs of the triarchic model.

The triarchic trait constructs are represented to varying degrees in alternative historical accounts of psychopathy (see Crego & Widiger 2016, Patrick 2018c) and different instruments for assessing it (Drislane & Patrick 2017, Patrick & Drislane 2015b, Sellbom 2018). **Supplemental Table 2** summarizes reported relations of various interview and self-report psychopathy inventories with scale measures of the triarchic traits. The triarchic trait constructs also connect with models of general personality including the five-factor model (FFM; Drislane et al. 2018, Poy et al. 2014) and Tellegen's (Tellegen & Waller 2008) Multidimensional Personality Questionnaire (MPQ) framework (Brislin et al. 2015). The representation of the triarchic traits in these established measures and models provides a link between their nomological networks and a means to integrate findings across studies. In addition, the triarchic model's explicit biobehavioral focus provides a basis for linking psychopathy to initiatives calling for greater use of physiological and behavioral measures to characterize and study psychopathology (Kozak & Cuthbert 2016, Kwako et al. 2016). Although the triarchic model was advanced as a framework for psychopathy, the biobehavioral processes corresponding to its constructs—threat sensitivity, affiliative capacity, and inhibitory control—are transdiagnostic (i.e., pertinent to clinical problems of many different kinds).

Issues of Definition and Measurement

One set of long-debated issues pertains to how psychopathy should be defined and measured. Issues of definition include whether psychopathy should be viewed as categorical (typological) or as dimensional (continuously varying) in nature and whether it is a unitary syndrome or a multifaceted condition. Measurement-related issues include questions of scope (What are the defining features of psychopathy? Should antisocial behavior items be included in diagnostic assessments?) and boundaries (What differentiates psychopathy from antisocial personality? Is boldness a component of psychopathy?). This section covers these issues and is followed by sections focusing on questions pertaining to the heterogeneity of psychopathy and its relationship to other mental disorders.

Categorical versus dimensional. As noted in the **Supplemental Appendix**, the field of clinical science has progressed toward a continuous dimensional approach to conceptualizing and assessing personality pathology, and Section III of the fifth edition of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5; APA 2013) includes a dimensional-trait system for diagnosing PDs as an alternative to the categorical system in Section II of the DSM-5. A questionnaire-based

Supplemental Material >

operationalization of the Section III alternative model, the Personality Inventory for DSM-5 (PID-5; Krueger et al. 2012), has been widely used since its inception (Barchi-Ferreira & Osório 2020, Watters & Bagby 2018).

Like other PDs, psychopathy was for many years investigated mainly in categorical terms—using ratings of resemblance to Cleckley’s (1976) clinical description as the method of assessment (for a review, see Hare 1978). The basis for a shift occurred with the introduction of the Psychopathy Checklist (PCL; Hare 1980) the same year the third edition of the DSM was published (DSM-III; APA 1980). Like the DSM-III, the PCL was a criterion-based system designed for clinical use (with correctional or forensic samples) that entailed the rating of examinees on 22 items selected to predict Cleckley-resemblance ratings. The availability of the PCL and its revised 20-item version (PCL-R; Hare 2003) led researchers to begin examining criterion variable associations for continuous psychopathy scores as well as cutoff-based groupings (e.g., psychopath versus nonpsychopath).

Subsequent studies using taxometric analysis revealed that psychopathy as assessed by the PCL-R and its variants is continuous (dimensional) rather than taxonic (Edens et al. 2006, Walters et al. 2007). These results, which converged with findings for DSM-defined PDs, were important because they encouraged the use of nonoffender samples in research as a means to understand the influences that contribute to psychopathy and its relations with various phenomena (e.g., other clinical conditions, laboratory measures, social-occupational functioning). Studies applying taxometric analysis to self-report psychopathy data for community participants, derived from inventories modeled directly after the PCL-R (e.g., Frick & Hare 2001, Levenson et al. 1995) or the broader psychopathy literature (e.g., Lilienfeld & Andrews 1996), likewise supported the view of psychopathy as dimensional (Edens et al. 2011, Walters et al. 2008). In line with this view, the triarchic model conceives of psychopathy in dimensional-trait terms.

Unitary versus multifaceted. Although psychopathy was conceptualized and studied for many years as a unitary condition presumed to have a discrete underlying cause (Cleckley 1976, Hare 1978, Lykken 1957, Patterson & Newman 1993), newer research has established that it encompasses separable symptom subdimensions with differing correlates. The best-established contemporary inventories for assessing psychopathy all contain factors or facets that reflect psychologically distinct subsets of symptoms. Even measures that were designed to index psychopathy as a unitary syndrome contain distinguishable (albeit correlated) factors.

For example, the interview-based PCL-R contains subsets of items that define interpersonal-affective (Factor 1) and impulsive-antisocial (Factor 2) subdimensions, each divisible into narrower facets (Hare et al. 2018). Inventories patterned after the PCL-R—including the informant-rated Antisocial Process Screening Device (APSD; Frick & Hare 2001) and self-report measures such as Paulhus et al.’s (2016) Self-Report Psychopathy Scale (SRP) and the Youth Psychopathic Traits Inventory (YPI; Andershed et al. 2002)—also contain correlated factors. However, subdimensions of psychopathy appear less correlated in certain assessment measures. The self-report Psychopathic Personality Inventory (PPI; Lilienfeld & Andrews 1996, Lilienfeld & Widows 2005), which indexes psychopathy in terms of traits represented in differing historical accounts, includes uncorrelated fearless dominance (FD) and impulsive antisociality [self-centered impulsivity (SCI)] factors, along with a narrower coldheartedness facet. Studies examining how subdimensions of different psychopathy inventories relate to measures of the triarchic model constructs (see **Supplemental Table 2**) have helped to clarify their similarities and differences.

Importantly, the subdimensions of psychopathy in different inventories show contrasting relations with external criterion measures. In some instances, correlations are specific to one subdimension or another (e.g., PCL-R Factor 2, but not Factor 1, relates to trait impulsiveness and

substance problems; Reardon et al. 2002). In others, opposing correlations are evident for one subdimension versus another (e.g., the PPI's FD factor relates negatively to trait anxiety and internalizing problems, whereas its SCI factor relates positively to these distress-related criteria; Benning et al. 2005a). The correlated subdimensions of measures such as the PCL-R or SRP show even stronger opposing relations with certain criterion measures when their shared variance is controlled for (e.g., measures of anxiousness, internalizing problems, and suicidal propensities; Hicks & Patrick 2006, Verona et al. 2001). This effect, known as statistical suppression (Paulhus et al. 2004), indicates the presence of separate constructs underlying these distinct subdimensions and has implications for understanding causal mechanisms for psychopathy.

Scope and boundaries. Multiple persisting debates center around what clinical features should be used to define (diagnose) psychopathy, which of these are more versus less central, and what other features should be viewed as concomitants or outcomes of psychopathy as opposed to defining elements. I consider these debates in relation to the numbered questions that follow.

Question 1: What are the defining features of psychopathy? An issue that scholars have wrestled with since the condition of psychopathy was first described by Pinel [1962 (1806)] is how broadly versus narrowly it should be defined. Pinel's characterization (termed *manie sans délire*, meaning "insanity without delirium") focused on reckless, destructive acts occurring without the disturbances in thought and speech that are typically seen in persons unable to control their behavior. For Pinel, the lack of behavioral restraint was the pathology, and the absence of "delirium" distinguished it from other types of mental illness. However, a problem with defining psychopathy in this way is that unrestrained nonnormative behavior can occur for various reasons in the absence of salient thought disorder. As a result, over the decades following Pinel, the term psychopathy came to encompass a broad range of behavioral conditions ranging from persistent criminality to substance addiction to sexual deviancy.

A major stated aim in Hervey Cleckley's (1976) influential book *The Mask of Sanity* (first edition published in 1941) was to counter this fractionation by characterizing psychopathy in terms of explicit diagnostic criteria that would identify a distinct clinical condition with a coherent etiological basis. Cleckley's criteria included ostensibly healthy features (social adeptness, coherent speech, lack of psychotic or anxious/neurotic symptoms, "immunity" to suicide) that masked a severe pathology marked by persistent reckless and impulsive behavior without regard for personal consequences or the feelings and welfare of others. One of the specific pathological features he listed was "inadequately motivated antisocial behavior," by which he meant acts such as "theft, forgery, adultery, [and] fraud [committed] in the absence of any apparent goal at all" (Cleckley 1976, p. 343). Cleckley viewed such behavior as part of a larger pattern of unrestrained deviancy that also included lack of planning, irresponsibility, impersonal sex, and a failure to learn from adverse experience.

From the standpoint of the triarchic model, the unrestrained deviancy that Cleckley (1976) characterized as the underlying pathology ("insanity") of psychopathy reflects externalizing proneness—the propensity toward impulse control problems of various types, which is evident in childhood as well as adulthood. However, Cleckley and other influential scholars (e.g., Karpman 1941, McCord & McCord 1964) maintained that impulsive-unrestrained behavior in itself does not constitute psychopathy. They viewed the presence of certain co-occurring clinical features to be exclusionary for this diagnosis. These included thought disorder [Pinel 1962 (1806)] or psychotic symptoms more broadly (Cleckley 1976), intellectual deficiency or obvious neurological dysfunction (Cleckley 1976, Karpman 1941), serious alcohol or drug addiction (Cleckley 1976), and anxious-depressive ("neurotic") symptomatology (Cleckley 1976, Karpman 1941, McCord & McCord 1964). The last of these exclusionary features is notable given that internalizing

symptoms systematically covary, to a moderate positive degree, with externalizing problems—at both younger (Achenbach & Edelbrock 1978) and older ages (Krueger 1999). The implication is that psychopathy represents a unique form of externalizing proneness that is distinct from what is found in cases that exhibit comorbid internalizing symptomatology.

Historic writers also identified characteristics specific to psychopathic individuals that distinguished them from other persons exhibiting salient externalizing behavior—in particular, untruthfulness and manipulateness, shallow emotions, absence of remorse, and a selfish disregard for the feelings and welfare of others. Cleckley (1976) considered these characteristics to be reflections of an underlying affective deficit that also contributed to the ostensibly adaptive, “mask” features of psychopathy—that is, social adeptness, imperviousness to stress, and disinclination toward self-harm. The triarchic model views these mask features instead as expressions of boldness (Patrick et al. 2009; see also Crego & Widiger 2016), a disposition rooted in low threat sensitivity that protects against internalizing but not externalizing problems. Consistent with this view, scales developed to index boldness correlate positively with interpersonal symptoms of psychopathy (Wall et al. 2015) and measures of social assertiveness and persuasiveness (Hall et al. 2004, Weiss et al. 2018) and negatively with measures of anxiousness/stress reactivity, internalizing symptoms, and suicidality (Benning et al. 2005a, Venables et al. 2015).¹

Drawing on this and other evidence, I (Patrick 2018c) posited that Cleckley’s conception of psychopathy, derived from experience with hospitalized psychiatric patients, reflects the confluence of two distinct dispositions—high externalizing proneness (disinhibition) and low fearfulness (boldness). To gain empirical insight into individuals of this type, my research team collected data from undergraduates and community adults prescreened for traits of disinhibition and boldness using the Triarchic Psychopathy Measure (TriPM; Patrick 2010) to ensure good representation of high and low scorers on each. PD symptoms were also assessed, using the Structured Clinical Interview for DSM-IV Axis II Disorders (SCID-II). **Supplemental Figure 1** shows mean symptom profiles for Cluster B and C PDs, along with depressive PD, for three subgroups of males from among those tested ($N = 100$): (a) low disinhibition/low boldness (bottom tercile on disinhibition, below median on boldness; $n = 10$), (b) high disinhibition/low boldness (top tercile on disinhibition, below median on boldness; $n = 15$), and (c) high disinhibition/high boldness (top tercile on both disinhibition and boldness; $n = 12$). As shown in **Supplemental Figure 1**, levels of Cluster B PD symptoms were elevated for both high-disinhibition groups relative to the low-disinhibition/low-boldness group (significantly for antisocial and borderline PD). However, the high-disinhibition/high-boldness group was distinguished from the high-disinhibition/low-boldness group by lower levels of Cluster C PDs (significant for avoidant and dependent PD) and depressive PD (also significant). Thus, the presence of high boldness mitigated disinhibition-related elevations on anxious-fearful and dysphoric symptomatology without dampening erratic-impulsive symptomatology.² Indeed, antisocial PD symptomatology was highest among subjects who exhibited high boldness together with high disinhibition.

Historic writers concerned with psychopathy in incarcerated criminal offenders (Craft 1966, Lindner 1944, McCord & McCord 1964) painted a more hostile/antagonistic picture in which

¹Of note, the personality and clinical symptom correlates of Factor 1 of the PCL-R, its interpersonal facet in particular (encompassing charm, grandiosity, lying, and manipulateness), parallel those of boldness—particularly when controlling for overlap with the PCL-R’s impulsive-antisocial (Factor 2) features (Douglas et al. 2008, Hall et al. 2004, Poythress et al. 2010).

²The high-disinhibition/low-boldness group also showed significantly higher levels of anxious-depressive (internalizing) symptoms, as assessed by the SCID-I.

features of shallow affect, absence of remorse, and selfish disregard were expressed in the form of callous exploitativeness and violent victimization. In the words of McCord & McCord (1964, p. 3): “[The psychopath] is an asocial, aggressive, highly impulsive person, who feels little or no guilt, and is unable to form lasting bonds of affection with other human beings.” The triarchic model uses the term meanness for these exploitative-aggressive features of psychopathy; researchers studying psychopathy in youthful samples (children, adolescents) have used the term callous-unemotional (CU) traits (Frick et al. 2014).

The triarchic model posits that the diagnostic features identified by Cleckley in psychopathic hospital patients, and those highlighted in descriptions of psychopathic criminals, reflect the constructs of disinhibition, boldness, and meanness. The reckless, unrestrained behavior emphasized in both—and in Pinel’s [1962 (1806)] original description—connects most strongly to the construct of externalizing proneness (also called disinhibition). However, neither Cleckley nor his criminological counterparts considered externalizing proneness in itself to be sufficient for a diagnosis of psychopathy. Additional features that reflected boldness (emphasized more so by Cleckley) or meanness (emphasized more in criminological accounts) were required. As described below in the section titled Heterogeneity of Psychopathy, the triarchic model proposes that differing configurations of these three dispositional characteristics can account for different observable expressions (subtypes or variants) of psychopathy.

Question 2: Should items pertaining to antisocial acts be included in assessments of psychopathy? Hare’s (2003) PCL-R was developed to assess for psychopathy, as described by Cleckley, in criminal offenders. Along with items pertaining to interpersonal and affective features, it contains a number of items that pertain to impulsive behavioral deviancy, including ones related to general propensities (e.g., impulsiveness, irresponsibility, lack of planning) and others that refer to more specific antisocial acts [e.g., early conduct problems, juvenile offenses, violations of conditional release, arrests and/or convictions for multiple types of offenses (“criminal versatility”)]. Youth psychopathy inventories patterned after the PCL-R (Frick & Hare 2001, Lynam 1997) also contain items that pertain to antisocial behavior and/or conduct problems (e.g., lying, stealing, illegal acts).

Cooke & Michie (2001) argued for removal of the PCL-R’s antisocial behavior items in characterizing the factor structure of the PCL-R, citing trait-oriented conceptions of PDs and the importance of distinguishing between basic tendencies (core personality traits) and characteristic adaptations (outward behavioral manifestations of traits). In a follow-up study, Cooke et al. (2004) evaluated the fit of two alternative models of the PCL-R—one in which its items loaded onto five subordinate factors that loaded in turn onto a single higher-order factor, and the other in which criminal behavior and relationship lability factors were modeled as causal outcomes of the PCL-R’s three trait-like factors (interpersonal, affective, impulsive-irresponsible; Cooke & Michie 2001). Fit was comparable for the two models, but the authors concluded on “logical grounds” that the latter model was preferable. However, a key methodological limitation of this study was that the data were cross-sectional (i.e., trait-like items and outcome items were collected together), precluding causal inferences (Neumann et al. 2006).

The question of whether items pertaining to criminal behavior or antisocial acts more broadly should be included as diagnostic criteria for psychopathy has been a continuing matter of debate (e.g., Hare & Neumann 2010, Hare et al. 2018, Skeem & Cooke 2010) and connects with broader discussions regarding how best to conceptualize relations between traits and clinical symptoms. Widiger (2011) considered three explanations: (a) Traits and symptoms mutually influence one another’s appearance or expression, (b) they exert causal effects on one another, and (c) they operate as coincicators of latent dimensions of psychological functioning. He reviewed evidence in

support of each of these possibilities and suggested that they are jointly operative and perhaps intertwined.

More recently, DeYoung et al. (2022) considered the association between traits and symptoms in relation to the HiTOP dimensional psychopathology model (Kotov et al. 2017). These authors argued, in line with Widiger's (2011) third explanatory concept, that traits and symptoms are coindicators of functionally relevant individual difference factors and that the major distinction between the two is the time frame of their occurrence (i.e., traits are expressed over longer periods of time, symptoms over shorter periods). DeYoung et al. further proposed that traits that are especially predictive of the occurrence of symptom states, termed maladaptive traits, covary with normative personality traits around shared latent dimensions, with the key difference being that maladaptive traits are indicative of extreme positions along those dimensions, at which individuals are likely to exhibit clinical dysfunction.

Vis-à-vis Widiger's (2011) other two explanatory concepts, DeYoung et al. (2022) posited that a causal relationship exists between traits and psychopathological symptoms that is best characterized in vulnerability terms. According to this view, individuals' positions along latent dispositional dimensions predispose them in varying degrees to the development of psychopathological symptoms. The relationship is probabilistic, in that the expression of higher dispositional vulnerability in terms of clinical symptoms is dependent upon the interplay of person factors and environmental influences across time. As such, this perspective acknowledges the importance of development to relations between traits and symptoms: Per Widiger's second explanatory point, traits can evolve over time as a function of accumulating experiences—including episodes of clinical dysfunction that heighten susceptibility to further problems in the future. A similar position was advanced by Clark (2005), who conceived of dispositional liabilities for psychopathology in temperament terms—highlighting advantages of this conceptualization for a biobehavioral-developmental analysis of pathways to mental illness.

Returning to whether items that pertain to criminal or antisocial acts should be included along with trait-like indicators in assessing for psychopathy, it is important to consider the purposes for which an assessment is to be used (Messick 1995). In research aimed at clarifying liability factors for psychopathy, delineating pathways from liability to symptomatic expression, and predicting the future occurrence of problems in asymptomatic individuals, the use of trait-like indicators is expected to be most beneficial. However, in assessments aimed at quantifying the presence and severity of pathology for outcome evaluations or clinical decision-making purposes (e.g., institutional assignment, treatment selection, readiness for release/discharge), strong grounds exist for the use of items pertaining to antisocial acts. This issue of applied-clinical utility connects to another long-standing topic of discussion—namely, the relationship between psychopathy as assessed by the PCL-R and DSM-defined antisocial personality disorder (ASPD).

Question 3: How different is PCL-R psychopathy from ASPD? ASPD as defined in Section II of the DSM-5 is a categorical diagnosis requiring symptoms of conduct disorder before age 15 (child symptoms) and the occurrence of further antisocial symptoms after this age (adult symptoms). This diagnosis has been criticized by psychopathy and PD researchers since its inception (see, e.g., Frances 1980, Hare et al. 1991). One concern is that ASPD has a relatively low diagnostic threshold with only a small number of child and adult symptoms (from an overall list of 22) required for its assignment. Given this, prevalence rates for ASPD in offender samples are so high [60–80% in incarcerated adult male samples (Hare 2003) versus ~3% among community males (APA 2000)] as to have limited utility in such samples. Another criticism is that ASPD emphasizes behavioral symptoms to the exclusion of affective-interpersonal features considered essential to the diagnosis

of psychopathy. As such, it has been viewed as phenotypically and etiologically diffuse compared to psychopathy.

For these and other reasons, the PCL-R has been favored for use in studies of psychopathy in adult offenders and for clinical use in correctional and forensic settings, parole/probation determinations, and judicial proceedings. It is considered an especially powerful predictor of institutional misconduct and potential for reoffending (Douglas et al. 2018). However, evidence exists that the construct assessed by the PCL-R as a whole is not so different from that indexed by the criteria for ASPD when quantified continuously (i.e., as a symptom-count variable) rather than categorically. In a sample of 74 male prisoners, Skilling et al. (2002) reported a correlation of 0.86 between total PCL-R scores (computed as the sum of 0–2 ratings for the PCL-R's 20 items) and symptom scores for DSM-IV ASPD (computed as the sum of 0–2 ratings for the adult criteria and a composite 0–2 rating for the child criteria as a whole; i.e., 8 items total). Assessments for the two measures were completed independently by different raters at separate points in time. Alpha reliability for the PCL-R items was 0.92; for the smaller number of ASPD items, it was 0.76. Of note, the ceiling for the correlation between these two measures, estimated as the square root of the product of their individual reliabilities, is 0.84. The implication is that continuous scores for the two index largely the same thing.

Consistent with the foregoing findings, Patrick et al. (2007) used structural equation analysis to model a general factor for the full PCL-R item set in 593 male offenders and reported a correlation of 0.92 between this latent PCL-R factor and DSM-IV ASPD symptomatology, computed as the mean of proportion scores (number met divided by total number) for the child and adult symptom criteria. Other findings from this study provide insight into the basis for the very high correlation between the two. Examination of the scree plot for the PCL-R item set revealed a dominant first factor with an eigenvalue (6.59) greatly exceeding those of subsequent factors (≤ 1.95)—indicative of a broad variance component saturating all items (Nandakumar 1993). The strongest indicators of this dominant broad factor (loadings > 0.5) were PCL-R items pertaining to antisocial deviance and criminal behavior—early conduct problems, juvenile delinquency, arrests and/or convictions for multiple types of offenses, irascibility/assaultiveness, callousness, fraudulence, boredom proneness, and irresponsibility.³ Four of the PCL-R's eight interpersonal-affective items—ones pertaining to charm, grandiosity, absence of remorse, and blame externalization—operated as weak indicators of this general factor (median loading = 0.25) and defined subsidiary factors distinct from the general factor.

Considered together with the findings of Skilling et al. (2002), these results indicate that the PCL-R as a whole effectively indexes the behavioral pathology (“insanity”; Cleckley 1976) of psychopathy as expressed in criminal form (McCord & McCord 1964). ASPD symptom scores appear to index this same pathology. It seems likely that this focus on behavioral pathology accounts for the effectiveness of the PCL-R in predicting clinical outcomes such as institutional misconduct and recidivism (for evidence of this, see Kennealy et al. 2010). In line with points made by DeYoung et al. (2022), the clinical utility of the PCL-R could likely be improved by adding items that assess for behavioral deviancy within a narrower, recent time frame (cf. Cooke & Logan 2018). This would allow for assessment of changes in symptom expression over time (e.g., as a function of aging, incarceration, and exposure to treatment) and would potentially enhance prediction of outcomes such as postdischarge adjustment and criminal recidivism.

³ Considering this evidence, it would have been more apt to use the label “Factor 1” for the impulsive-antisocial features of the PCL-R rather than for the interpersonal-affective features.

Other findings reported by Patrick et al. (2007) indicate that the PCL-R assesses psychopathy-related traits secondarily to, and as intertwined with, behavioral pathology. This is problematic because the pathology-related variance in the PCL-R's dispositional items operates to obscure their relations with outcomes of interest—owing to statistical suppression, as described above. This complication is documented by many studies that have reported stronger and more distinctive relations of the PCL-R's two broad factors and three trait-like facets with criterion measures when controlling for their shared variance (for reviews, see Patrick & Bernat 2009, Patrick & Drislane 2015a). Vis-à-vis arguments for the importance of traits to an understanding of psychopathology, clearer separation between behavioral symptoms and traits in the assessment of psychopathy is desirable—particularly in etiological and developmental research. This leads to another topic of continuing debate: the relevance of boldness to psychopathy (e.g., Lilienfeld et al. 2016, Miller & Lynam 2012).

Question 4: Is boldness a component of psychopathy? As discussed above, impulsive externalizing behavior is highly characteristic of clinically psychopathic individuals, but persons high in externalizing differ as a whole from individuals described as psychopathic by Cleckley (1976) and others (Karpman 1948, McCord & McCord 1964). High externalizing is generally associated with (a) increased rather than decreased internalizing symptomatology (Achenbach & Edelbrock 1978, Krueger 1999), (b) higher self-ratings of anxiety, neuroticism, and negative affectivity (e.g., Ellingson et al. 2018, Sher & Trull 1994), and (c) elevated risk for suicidal ideation and action (Verona & Patrick 2000). From this standpoint, the psychopathic cases Cleckley documented are markedly unusual: They exhibited severe externalizing behavior without accompanying distress or internalizing problems, and (in Cleckley's words) they showed a "specific and characteristic immunity" to suicide (p. 359).

The triarchic model accounts for this symptom pattern by characterizing Cleckley's cases as high in boldness as well as disinhibition—with the former reflected in the mask features and the latter in the behavioral deviancy features. Consistent with this view, Crego & Widiger (2016) found that raters unfamiliar with the concept of boldness judged Cleckley's cases to be very high in traits related to it. Other work has shown that the prototypic FFM personality profile for psychopathy identified by Miller et al. (2001) includes salient representation of boldness: Studies operationalizing boldness using scores on the PPI's FD factor (Ross et al. 2009) or the TriPM boldness scale (Poy et al. 2014) have reported correlations of 0.5 to 0.6 with psychopathy prototype scores computed from the NEO-PI-R personality inventory. In addition, research has shown that boldness is represented in many prominent assessment inventories for psychopathy, including the PCL-R and its self-report counterpart (the SRP), the PPI, and the YPI (Drislane et al. 2014, Venables et al. 2014; see **Supplemental Table 2**). Of note, two separate studies (Venables et al. 2014, Wall et al. 2015) have shown that the representation of boldness within the PCL-R—in its interpersonal symptom facet in particular (Benning et al. 2005a)—is what differentiates PCL-R psychopathy most from ASPD.

Although the interpersonal facet of the PCL-R includes elements of boldness, it correlates to a moderate positive degree with its impulsive and antisocial facets. As a result, PCL-R interpersonal facet scores do not correlate inversely as expected with measures of anxiousness/negative affectivity (Hall et al. 2004) or suicidality (Douglas et al. 2008) unless the variance they share with the PCL-R's impulsive-antisocial items is controlled for. By contrast, boldness in the triarchic model is conceptualized as independent of disinhibition, and scales designed to index boldness show negative zero-order associations with criterion measures of anxiousness, negative affectivity, internalizing symptomatology, and suicidality (Douglas et al. 2008, Patrick 2018c, Sellbom 2018, Venables et al. 2015). However, because boldness is independent of rather than inversely

related to disinhibition, the two attributes can readily co-occur. When they do, boldness appears to amplify certain deviant propensities while at the same time operating to mitigate internalizing symptomatology—as illustrated by the diagnostic profiles of high- versus low-boldness subgroups of high-disinhibition individuals described above and depicted in **Supplemental Figure 1**.⁴

Heterogeneity of Psychopathy

A topic of continuing interest in the literature is the existence of different subtypes of psychopathy that result from differing causal factors. This notion is reconcilable with a dimensional-trait approach to psychopathology if one presumes that particular configurations of traits can yield subgroups of individuals with clinically salient features of one sort or another (Lilienfeld et al. 2019). It is also compatible with the developmental principle of multifinality (Cicchetti & Rogosh 1996), which holds that phenotypically similar outcomes can arise from different causal influences.

One form of psychopathy is the “masked insanity” type described by Cleckley (1976), which according to the triarchic model reflects the conjunction of high boldness and high disinhibition. However, Cleckley’s description was based on experiences with psychiatric inpatients who engaged mainly in nonviolent externalizing acts, for unknown reasons; in his time, such individuals were often institutionalized in psychiatric hospitals rather than prisons. As noted above, an alternative characterization of psychopathy, emphasizing predatory aggressive deviancy that often involves coldhearted acts of violence, emerged out of research with incarcerated offenders. From a triarchic model perspective, this variant entails high meanness (callousness-unemotionality) along with high disinhibition. In contrast to Cleckley’s inpatient cases, individuals of this type appear cold, uncaring, and antagonistic rather than pleasant, alert, amiable, and easy to talk with.

In addition to masked and predatory criminal variants, another subtype described in the historic literature is the secondary psychopath (Karpman 1941, 1948). This label has generally been applied to impulsive-antisocial individuals who are high in anxious-neurotic tendencies; it is assumed that the behavioral deviancy in such cases is an expression of inner conflict engendered by adverse life experiences. However, an alternative view is that individuals of this type are high in disinhibition, but not in boldness or callousness-unemotionality. As noted above, disinhibition (general externalizing proneness) is substantially heritable (Krueger et al. 2002, Yancey et al. 2013) and positively correlated with anxious-neurotic tendencies. As such, high disinhibition in itself appears sufficient to account for what has been called secondary psychopathy without the assumption of a unique environmentally based etiology.

Given that disinhibition and meanness are moderately correlated, violent criminal offenders with elevated scores on both are expected to include a mix of individuals, some exhibiting aggressive behavior due to anger and weak restraint associated with disinhibition, and others exhibiting aggression more as a function of emotional insensitivity, low social concern, and predatory goal-seeking. Consistent with these expectations, research has shown that adult offenders who score as psychopathic on the PCL-R comprise two subgroups with contrasting personality profiles—one

⁴A key question is whether individuals jointly high in boldness and disinhibition would also exhibit the “affective” symptoms (lack of remorse or shame, poverty in affective reactions, inability to love, lack of social reciprocity) described by Cleckley (1976), which appear most related to meanness. Elsewhere, I (Patrick 2018c) suggested—based on the FFM correlates of boldness (Poy et al. 2014) and research modeling externalizing traits and symptoms (Krueger et al. 2007)—that persons high in boldness as well as disinhibition would likely be untroubled by their actions, lacking in empathic concern, untrustworthy, and disloyal. Consistent with this hypothesis, high-boldness/high-disinhibition participants in the diagnostic study for which results are depicted in **Supplemental Figure 1** showed elevated scores on TriPM Meanness relative to the low-boldness/low-disinhibition group, whereas participants high only in disinhibition did not.

marked by very low anxiousness and an active (agentic) social style, and the other by very high hostility/aggressiveness along with heightened anxiety and impulsiveness and low social affiliation (Hicks et al. 2004, Skeem et al. 2007). It seems likely that offenders high in boldness as well as callous-disinhibitory tendencies fall mainly into the first of these subgroups, whereas the latter subgroup includes offenders high in disinhibition and meanness (but not boldness) along with some primarily high in disinhibition. Cluster analytic studies of adults assessed using self-report psychopathy measures (e.g., Drislane et al. 2015, Falkenbach et al. 2014) have likewise revealed distinct primary and secondary subtypes, the former marked by low anxiousness and high social assertiveness along with antisocial deviance, and the latter by high anxiousness and internalizing symptomatology along with alienation, hostility, aggressiveness, and substance problems.

Turning to the child psychopathy literature, substantial evidence indicates that youth with conduct problems who display CU traits, compared to those who lack these traits, show a more severe pattern of antisocial behavior that involves proactive as well as reactive aggression (Frick & Marsee 2018, Frick et al. 2014). This evidence served as the impetus for inclusion of a specifier in the DSM-5 to distinguish variants of conduct disorder with and without CU traits. However, recent research has documented distinct subgroups of youth that exhibit conduct problems with CU traits and that can be distinguished by low versus high anxiousness (Fanti et al. 2013) and perhaps as well by history of trauma exposure (Docherty et al. 2016, Kahn et al. 2013). Other recent research has begun to examine the role of traits of disinhibition and boldness, along with callousness-unemotionality, in the development (Bertoldi et al. 2021, Brislin et al. 2019) and expression (e.g., Sica et al. 2020, Somma et al. 2016) of antisocial behavior in younger-aged individuals.

An important direction for future research will be to investigate the behavioral manifestations of differing configurations of psychopathy-related traits at particular ages and across development. In addition to comparing high-boldness/high-disinhibition, high-meanness/high-disinhibition, and high-disinhibition-only individuals, participants who represent other configurations of triarchic traits will be interesting to recruit and study. For example, low-disinhibition individuals who score high on boldness, or on boldness as well as meanness, may be likely to exhibit characteristics of “successful” psychopathy (Benning et al. 2018, Kranefeld & Bickel 2021). Individuals low in disinhibition and boldness but high in meanness will also be interesting to investigate, particularly in light of work suggesting that meanness in itself may dispose to behavioral deviance of certain severe types (e.g., serial murder; Hickey et al. 2018).

Psychopathy and Other Mental Disorders

Traditionally, psychopathy has been viewed as a discrete psychopathological syndrome with a distinct underlying cause, and there has been a longstanding inclination to study it unto itself rather than in relation to other mental disorders. The most widely used assessment measures for psychopathy were developed separately from the DSM diagnostic system, and prominent theories for explaining it (Patrick 2018b) are largely psychopathy-specific. However, psychopathy shows systematic relations with other forms of psychopathology. As noted above, when defined in terms of overall scores on the PCL-R, which disproportionately reflect impulsive-antisocial proclivities, psychopathy relates very strongly to ASPD as defined in the DSM. Other DSM-defined conditions that show robust relations with psychopathy as assessed by the PCL-R and alternative inventories are substance use disorders and narcissistic PD (Widiger & Crego 2018). Other work by Krueger et al. (2007) on the latent structure of externalizing problems and traits has revealed a general disinhibitory factor along with residual callous-aggressive and substance abuse factors. The general externalizing and callous-aggressive factors of this model served as referents—along with findings from the child psychopathy and personality/psychopathology literatures—for the triarchic constructs of disinhibition and meanness (Patrick et al. 2009). From an externalizing

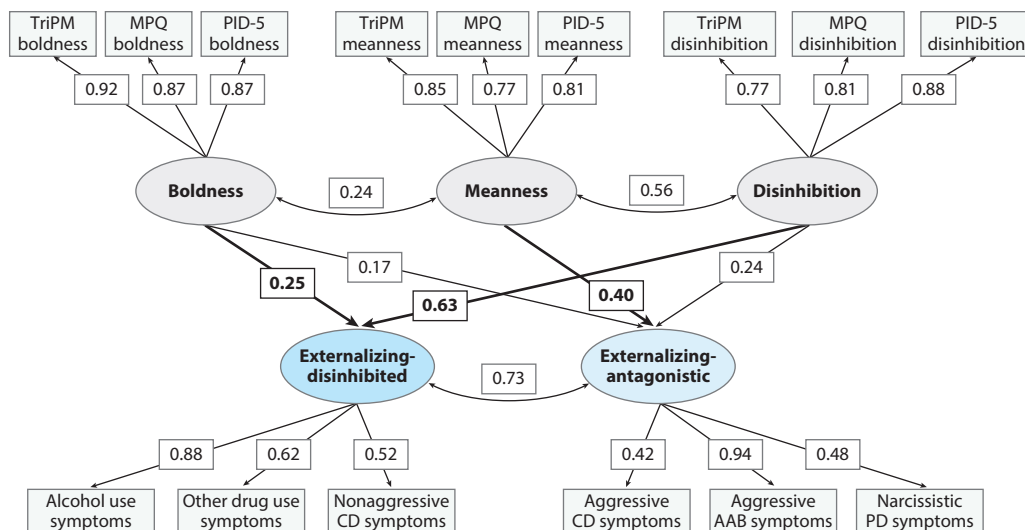


Figure 1

Latent-variable model of relations between triarchic trait dimensions and subdimensions of externalizing psychopathology (disinhibited, antagonistic) ($N = 212$). Coefficients for paths between triarchic trait factors and externalizing factors are beta values from regression models using the three traits together as predictors of each externalizing factor ($p < 0.05$ and < 0.01 , respectively, for unbolded and bolded coefficients). Model $\chi^2(80) = 191.02$, $p < 0.001$; root mean square error of approximation = 0.085; comparative fit index = 0.920; Tucker–Lewis index = 0.895; standardized root mean squared residual = 0.057. Abbreviations: AAB, adult antisocial behavior; CD, conduct disorder; DSM-5, *Diagnostic and Statistical Manual of Mental Disorders*, Fifth Edition; MPQ, Multidimensional Personality Questionnaire; PD, personality disorder; PID-5, Personality Inventory for DSM-5; TriPM, Triarchic Psychopathy Measure.

model perspective, substance problems relate to psychopathy via its disinhibition facet, but they are distinct from its meanness facet.

As an empirical illustration, **Figure 1** depicts a structural model characterizing relations of correlated disinhibited and antagonistic externalizing factors of DSM psychopathology (Kotov et al. 2017, Krueger et al. 2021) with the triarchic trait dimensions. The data are from a mixed-gender sample of undergraduates and community adults ($N = 212$); the subgroups of males differing in traits of boldness and disinhibition discussed above (and depicted in **Supplemental Figure 1**) were drawn from this same sample. The DSM-IV symptom variables used as indicators of the two externalizing psychopathology factors were derived from clinical interview (SCID-I and SCID-II) assessments. The indicators of the triarchic trait dimensions consist of different self-report scale measures of these traits (Brislin et al. 2015, Drislane et al. 2019, Patrick 2010; see **Supplemental Table 1**). In the model, the two externalizing factors were regressed onto the latent triarchic traits to evaluate distinct relations for the traits as concurrent predictors. As shown, the model as a whole exhibited acceptable fit. Disinhibited externalizing, defined by alcohol and other drug use symptoms along with nonaggressive conduct disorder symptoms, was predicted strongly by triarchic disinhibition (B coefficient = 0.63) and to a lesser unique extent by boldness ($B = 0.25$), with meanness predicting negligibly. By contrast, antagonistic externalizing, defined by aggressive child and adult symptoms of ASPD along with narcissistic PD symptoms, was predicted most strongly by triarchic meanness ($B = 0.40$) and to lesser degrees by disinhibition and boldness (B coefficients = 0.24 and 0.17).

Regarding anxious-depressive psychopathology, Cleckley (1976) characterized psychopathic patients as specifically lacking in nervousness and free from “psychoneurotic” symptoms.

Supplemental Material >

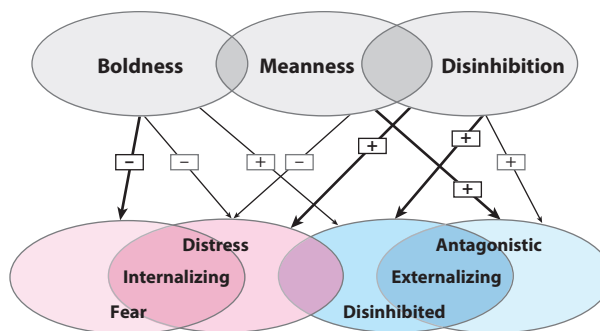


Figure 2

Schematic depiction of regression model relations for latent triarchic traits as joint predictors of subdimensions of internalizing and externalizing psychopathology, based on findings reported by Latzman et al. (2020) and those depicted in **Figure 1**. Unbolded plus sign or minus sign indicates that the beta-coefficient equals or exceeds $|0.2|$ but not $|0.4|$; bolded plus sign or minus sign indicates that the beta-coefficient equals or exceeds $|0.4|$.

Consistent with this historic characterization, contemporary researchers and clinicians with expertise in psychopathy have consistently identified low anxiousness and lack of emotional vulnerability as typical of psychopathic individuals (Kreis et al. 2012, Miller et al. 2001). However, by contrast, overall scores on the PCL-R and other PCL-based inventories appear largely unrelated to measures of anxiety and internalizing problems (Hare 2003, Schmitt & Newman 1999, Widiger & Crego 2018). As discussed above, these theory-inconsistent findings can be explained by the strong representation of impulsive-antisocial deviancy in these measures, which operates to suppress expected negative relations for interpersonal-affective features. This was demonstrated empirically by Hicks & Patrick (2006), who reported stronger opposing associations for the PCL-R's two broad factors with latent fear and distress factors (modeled using self-report scale indicators of each) when controlling for their shared variance, relative to associations observed for each PCL-R factor in toto.

More recently, Latzman et al. (2020) characterized relations between latent-variable representations of the triarchic traits and fear versus distress factors of internalizing psychopathology (Kotov et al. 2017) modeled using SCID-assessed symptom scores as indicators. The participant sample for this work was the same as that employed in the above-described model for disinhibitory and antagonistic factors of externalizing, and the same self-report scales were used to index the triarchic trait dimensions. Indicators of the fear-disorder factor of internalizing were symptom scores for specific phobia, social phobia, agoraphobia, and panic disorder; indicators of the dysphoria/distress factor were symptom scores for major depressive disorders, dysthymia, and post-traumatic stress disorder. As in the model for externalizing psychopathology, unique predictive relations for the latent triarchic traits were evaluated via regression. The fear factor of internalizing showed a strong negative association with triarchic boldness ($B = -0.65$), a weaker positive relationship with disinhibition ($B = 0.21$), and a negligible association with meanness. By contrast, the dysphoric/distress factor showed comparable negative relations with boldness and meanness (B coefficients = -0.26 and -0.24) and a strong positive association with disinhibition ($B = 0.58$).

These findings for internalizing together with those for externalizing are summarized schematically in **Figure 2** and illustrate how traits that relate to distinct symptom facets of PCL-R psychopathy (per **Supplemental Table 2**) can be linked to broad dimensions of the HiTOP

psychopathology framework. Of note, the alternative model for personality disorders (AMPD) in the DSM-5 is organized around maladaptive trait domains that resemble higher-order dimensions of the HiTOP framework (see Kotov et al. 2017), and it includes trait-based characterizations of ASPD and psychopathy that mesh with the triarchic-trait/externalizing model depicted in **Figure 2**. Specifically, the AMPD defines antisocial personality in terms of traits from domains of disinhibition and antagonism, and [in line with the findings of Venables et al. (2014) and Wall et al. (2015)] it includes a psychopathic features specifier encompassing traits related to boldness (Strickland et al. 2013).

However, the disinhibition and antagonism domains of the AMPD do not correspond directly to dispositional dimensions of disinhibition and meanness in the triarchic model (Strickland et al. 2013); therefore, it is necessary to configure lower-level AMPD traits differently to approximate triarchic disinhibition and meanness more closely (Drislane et al. 2019; see also Bowyer et al. 2020). Likewise, triarchic boldness does not map clearly onto any one domain of the AMPD and is best indexed by a combination of traits from different AMPD domains (Drislane et al. 2019, Strickland et al. 2013). This is also true of the normative FFM counterparts to the AMPD traits: FFM conscientiousness and agreeableness do not relate clearly to triarchic disinhibition and meanness, and lower-level traits from different FFM domains need to be aggregated to index boldness (Drislane et al. 2018, Poy et al. 2014). However, this point raises the following question: Why configure lower-level traits along triarchic model lines as opposed to AMPD or FFM lines? The answer, addressed in the next two sections, is that the triarchic model traits are conceptualized in biobehavioral-developmental terms (Patrick et al. 2009), were deduced in part from biobehavioral research findings, and, owing to ongoing research, have nomological networks that include variables from behavioral and neurophysiological as well as self-report and clinical-diagnostic modalities (see **Supplemental Table 3**). Importantly, as discussed below under the section titled Neurobiological Research, the triarchic traits are viewed as open constructs that are subject to revision based on accumulating knowledge of their nomological networks.

Supplemental Material >

UNDERSTANDING PSYCHOPATHY IN BIOBEHAVIORAL-DEVELOPMENTAL TERMS

Along with issues of conceptualization/assessment and phenotypic subtyping, contemporary research on psychopathy has been concerned with understanding this condition in biobehavioral and developmental terms. Detailed coverage of the existing literature in this area is beyond the scope of this article, so the reader is directed to recent reviews for in-depth summaries (see, e.g., Blair et al. 2018, Frick & Marsee 2018, Patrick et al. 2018b, Yang & Raine 2018). My aim in this section is to provide a brief overview of major ideas and findings, with reference to the triarchic model, as a basis for considering future research directions, as discussed in the last major section, titled Conclusions and Recommendations for Further Ongoing Research.

Biobehaviorally oriented research on psychopathy in adult offenders focused for many years on two alternative theoretical models. One of these posited that psychopathic symptoms arise from a weakness in emotional sensitivity (Cleckley 1976), particularly in the capacity for anxiety or fear (Hare 1965, Fowles 1980, Lykken 1957). The other hypothesized that psychopathic symptoms result from a deficit in cognitive-attentional processing that impairs inhibitory control and punishment learning (Patterson & Newman 1993). Following identification of distinct symptom components (factors) in PCL-R psychopathy, researchers sought to examine relations of these symptom components with biological and behavioral measures. Work of this kind yielded evidence for impaired emotional response (to aversive cues, especially) being preferentially related to the affective-interpersonal (Factor 1) component (e.g., Patrick 1994) and for impaired cognitive-attentional

processing being related more to the impulsive-antisocial component (Morgan & Lilienfeld 2000, Moltó et al. 2007).

As noted above, research in the 2000s employing the PPI to investigate psychopathy in nonoffenders (Benning et al. 2003) revealed distinct fearless-dominant and impulsive-antisocial factors akin to those of the PCL-R, but uncorrelated rather than correlated. Similarities in the psychological (e.g., Benning et al. 2005a) and physiological (e.g., Benning et al. 2005b) correlates of PPI FD to those of PCL-R Factor 1 led to the triarchic concept of boldness, a disposition rooted in low threat sensitivity that contributes to the interpersonal-affective features of psychopathy. Commonalities were also evident between the psychological (e.g., Benning et al. 2003) and physiological (e.g., Carlson et al. 2009) correlates of PPI impulsive antisociality and those of general externalizing proneness (Krueger 1999, Krueger et al. 2021, Patrick et al. 2006). This evidence, together with research documenting a close link between PCL-R Factor 2 and externalizing proneness (Patrick et al. 2005), led to the triarchic concept of disinhibition, a disposition rooted in weak frontal-inhibitory control that contributes to the impulsive-antisocial features of psychopathy. These lines of research served as the basis for a two-process (Patrick & Bernat 2009) or dual-process (Fowles & Dindo 2009) conceptualization of psychopathic behavior.

Contemporary research on psychopathy in youth, beginning in the 1990s, has focused in particular on the idea of a distinct variant of conduct disorder marked by the presence of CU traits. This variant is associated with especially severe and aggressive behavioral deviancy that is more likely to persist into adolescence and adulthood and lead to criminal justice involvement (Frick & Marsee 2018, Frick et al. 2014). It is believed to have a distinct biobehavioral basis and require specialized methods of prevention and treatment relative to other variants of conduct disorder. Initially, biobehavioral theory and research on CU traits focused on deficits in sensitivity to threat and distress cues and on the role of the amygdala in these deficits (Blair 2006, Marsh et al. 2008, Moul et al. 2012). In an effort to reconcile this view of callousness-unemotionality with evidence for a prominent role of weak threat sensitivity in the boldness facet of psychopathy (for a recent review, see Patrick 2018a), the triarchic model posited that low affiliative capacity contributes to a CU (mean) as opposed to a bold expression of weak threat sensitivity. Recent writings on psychopathy in youth have begun to consider the role of low affiliation, as distinguished from threat sensitivity, in callousness-unemotionality (Viding & McCrory 2019, Waller & Wagner 2019).

Until recently, the literature on psychopathy in youth has not considered dispositional boldness or disinhibition in relation to the emergence, expression, and developmental course of psychopathic symptomatology. This has begun to change through new studies administering established triarchic scale measures such as the TriPM to adolescent samples (e.g., Kyranides et al. 2017, Sica et al. 2020, Somma et al. 2016) and through work being done to create triarchic scale measures from questionnaire items included in already-existing longitudinal-developmental data sets (e.g., Bertoldi et al. 2021, Palumbo et al. 2021). Reciprocally, efforts are being made to bridge between the youth and adult literatures by evaluating whether known behavioral and brain correlates of CU traits such as impaired recognition and deficient neural response to fearful face stimuli extend to meanness as assessed in adults (Brislin & Patrick 2019, Brislin et al. 2018). Work is also being done to test for affiliative deficits in relation to this facet of psychopathy in adults (Mejia et al. 2020, Vieira & Marsh 2014).

Despite these advances, biobehavioral-developmental understanding has been hampered by diversity in the measures used to assess psychopathy within and across age groups and by broad variability in explanatory concepts and methods for operationalizing them. These issues create problems for interpreting results from individual studies and integrating findings across studies. In the next, final section, I describe a systematic strategy for coordinating biobehavioral-developmental

research on psychopathy that can allow it to be informed by, and in turn help to inform, ongoing research on mental health problems more broadly.

CONCLUSIONS AND RECOMMENDATIONS FOR FURTHER ONGOING RESEARCH

Conceptualization and Assessment

Available evidence indicates that psychopathy is continuous and multidimensional. It can be characterized either in clinical-symptom or trait-dispositional terms. The dominant instrument for assessing psychopathic symptomatology in adult criminal offenders over the past 30 years has been the PCL-R. Total scores on the PCL-R substantially reflect engagement in illicit behaviors at younger and older ages and correlate very highly with continuous ASPD symptom scores, which reflect aggressive and nonaggressive antisocial acts before and after age 15. From a HiTOP model perspective, the clinical-symptom dimension assessed in common by the PCL-R and by the criteria for ASPD reflects antagonistic externalizing. A profitable direction for future research would be to use item response modeling to integrate and refine the PCL-R and ASPD criterion sets into an optimally efficient and effective measure for assessing antagonistic externalizing symptomatology in correctional and forensic samples. Assessment of broad externalizing and its subdimensions according to the HiTOP model can provide a means for indexing the counterpart behavioral pathology of psychopathy in nonoffender samples.

From a personality trait standpoint, not all offenders who score very high on the PCL-R as a whole are the same. Two distinct subgroups have been identified: one low in anxiousness and high in agency/assertiveness, the other high in anxiousness and irritability/aggressiveness (Hicks & Drislane 2018). The former group (labeled primary or stable) may resemble more the masked externalizing variant of psychopathy described by Cleckley (1976) in psychiatric inpatients. The latter group (labeled secondary or aggressive) may resemble more the life-course-persistent antisocial individual described by Robins (1966) and Moffitt (1993). The DSM-5 AMPD accounts for this distinction through inclusion of a psychopathy specifier for ASPD, reflecting traits indicative of the primary variant. The triarchic model posits that distinct variants of clinical psychopathy reflect the presence of boldness versus meanness in conjunction with general externalizing proneness (disinhibition). It will be valuable in future research to further examine the characteristics of individuals with extreme scores on different combinations of psychopathy-related traits as specified by the triarchic model or the AMPD.

Developmental Research

Contemporary research on psychopathy in clinic-referred youth has focused on characterizing a distinct subgroup that exhibits conduct problems accompanied by CU traits. Conduct problems have been operationalized in various ways in younger-aged samples, including through use of DSM conduct disorder diagnoses or symptom counts, variants of Achenbach's (2009) clinical assessment inventory, and items from youth psychopathy inventories such as the APSD. This creates difficulty for integrating results across studies. A further obstacle is the use of different measurement modalities (informant report, self-report, clinical interview) to assess conduct problems at younger versus older ages.

An important goal for future research will be to refine and coordinate dimensional assessments of antisocial deviance in youth to permit better integration of findings within and across age levels. In doing so, efforts should be made to improve quantification of aggressive and nonaggressive symptom facets, given evidence for differences in their correlates and etiological bases (Burt 2009,

Kendler et al. 2013) and the potential they have to interface with externalizing subdimensions of the HiTOP model (see **Figure 1**).

Research on psychopathy in youth has only recently begun to examine how the triarchic traits of disinhibition and boldness contribute to the emergence and expression of conduct problems, relative to and in concert with callousness-unemotionality (meanness). An initial study using the TriPM with adolescents aged 16–17 years (Somma et al. 2016) reported similar-level unique associations for all three triarchic traits when entered together as predictors of self-reported engagement in delinquent acts over the preceding year. Another study of 14- to 18-year-olds that used a shortened (half-length) version of the TriPM (Sica et al. 2020) reported significant unique effects for disinhibition and meanness, but not boldness, in predicting self-reported conduct problems over the preceding 6 months. Boldness in this study was related, however, to lower reports of negative emotional states and less susceptibility to anxious/depressive symptoms.

An alternative to using established inventories like the TriPM is to create scale measures of the triarchic traits using construct-relevant items from questionnaire measures included in already existing data sets. Employing this approach to assess the triarchic traits in a longitudinal sample, and using regression analysis to test for unique contributions of each trait to prediction of clinical outcomes, Bertoldi et al. (2021) reported positive relations for boldness and disinhibition (but not meanness) assessed at age 14–15 with reported engagement in nonaggressive antisocial acts at age 19–20; meanness at age 14–15, by contrast, emerged as the sole unique predictor of reported aggressive acts at age 19–20. In contrast with their joint positive prediction of later nonaggressive antisociality, boldness and disinhibition at age 14–15 contributed in opposing directions (negative and positive, respectively) to prediction of reported internalizing symptomatology at age 19–20. These longitudinal study results provide evidence for the triarchic traits as indicators of liability versus immunity to problems of different types and align with the cross-sectional findings shown in **Figure 2** for relations of latent triarchic traits with HiTOP psychopathology dimensions in adults.

Neurobiological Research

In thinking about the role of neurobiological systems and processes in psychopathy and other forms of mental illness, it is helpful to consider relations among biology, traits, and symptoms in ontogenetic-developmental terms (Patrick & Hajcak 2016, Perkins et al. 2020). As noted above, symptoms and traits can be viewed as coindicators of latent person-factors with differing time frames—symptoms generally being expressed over shorter intervals, traits over longer periods (DeYoung et al. 2020). Normative traits and maladaptive traits each relate probabilistically to symptom states, but maladaptive traits include dysfunctional elements and are thus more strongly predictive of symptom occurrence. Although more stable than symptoms, traits are influenced by ongoing maturational and environmental factors across time and are thus susceptible to change, meaning that dispositional susceptibility to clinical dysfunction in one form or another can change across time as well.

Variations along psychological trait dimensions have long been viewed as intertwined with variations in neural function (Allport 1937, Eysenck 1967). However, observed relations between trait scores and indicators of neural function tend to be small and inconsistent because of conceptual and measurement mismatch. **Supplemental Figure 2** illustrates a strategy for interfacing neural systems and processes with clinical-symptom scores, involving the use of multimodal operationalizations of biobehavioral trait constructs; the two traits depicted in the figure, threat sensitivity and inhibitory control, are biobehavioral counterparts to boldness and disinhibition in the triarchic model. Expanding upon this illustration, **Figure 3** depicts different-modality indicators of biobehavioral traits corresponding to all three triarchic constructs, within a framework

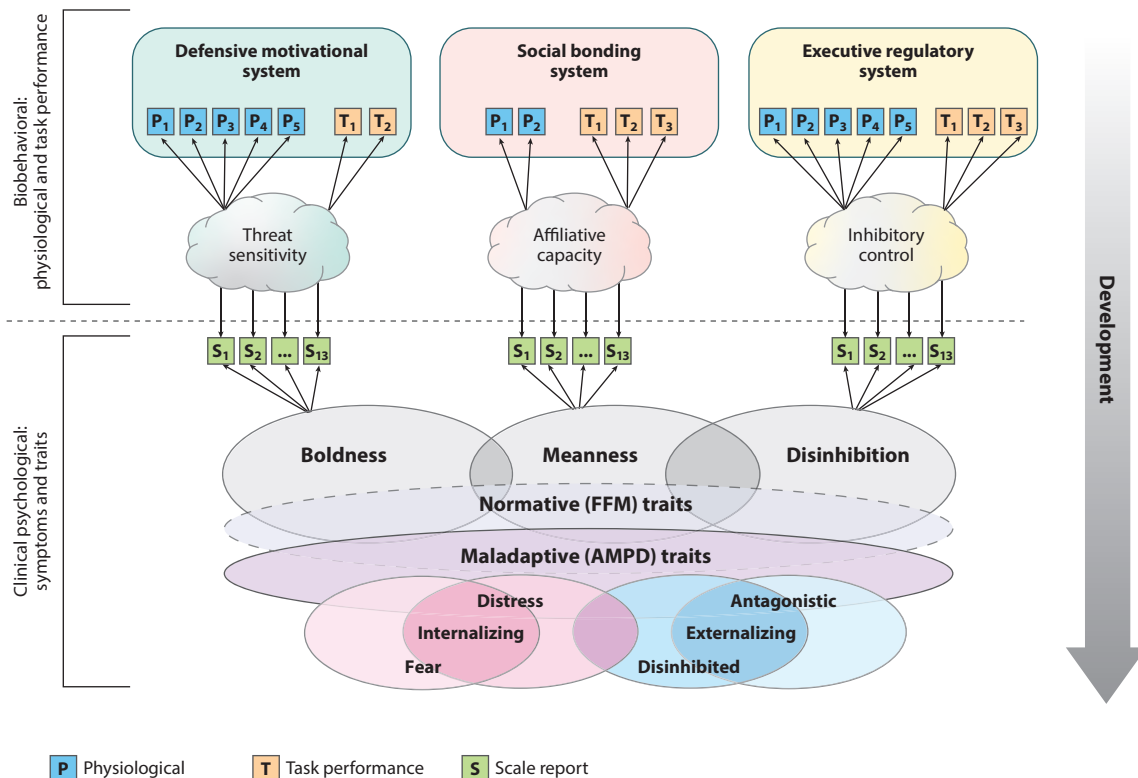


Figure 3

Schematic depiction of a conceptual-empirical framework for interfacing neural systems constructs (defensive motivational, social bonding, and executive regulatory; upper level) with trait and symptom dimensions (lower level), through use of multimodal representations of biobehavioral traits of threat sensitivity, affiliative capacity, and inhibitory control (middle level). Ovals denote trait or symptom dimensions that can be modeled as latent variables using report-based indicators (scale scores, clinical ratings) with documented covariance patterns. Cloud shapes denote hypothesized biobehavioral trait dimensions for which provisional indicators from different modalities have been identified (for listings of these, see **Supplemental Tables 1 and 3**) but whose covariance patterns need to be fully delineated to establish effective latent-variable models. Indicators of target biobehavioral traits from the self-report modality operate also as indicators of latent triarchic factors that link directly to normative (Drislane et al. 2018, Poy et al. 2014) and maladaptive (Drislane et al. 2019, Strickland et al. 2013) traits and in turn to internalizing and externalizing psychopathology dimensions. Biobehavioral trait indicators from the physiological and task-performance modalities link directly to neural systems constructs. Conceptualizations of biobehavioral traits will be shaped by accumulating knowledge of covariance patterns among replicable indicators of these traits from different measurement modalities. The right-side arrow denotes the important role of development in this framework. Abbreviations: AMPD, alternative model for personality disorders; FFM, five-factor model.

for interfacing neural systems with normative-FFM and maladaptive-AMPD traits, and in turn with HiTOP psychopathology dimensions.

A more detailed discussion of the framework depicted in **Figure 3** can be found in other published works (e.g., Patrick & Drislane 2015b; Patrick et al. 2012, 2013, 2019a). However, some specific points warrant mention. One is that multiple scale sets exist for operationalizing biobehavioral traits corresponding to boldness, disinhibition, and meanness in the modality of self-report (denoted in **Figure 3** by squares labeled with subscript-numbered S's). As shown in **Supplemental Table 1**, some of these scale sets are composed of items from specific psychopathy or omnibus personality inventories, and others consist of items taken from different questionnaire measures

Supplemental Material >

included in specialized existing data sets. These various scale sets have been created using a common construct-based development protocol (see Hall et al. 2014) and validated against one another and in relation to pertinent criterion variables (including clinical outcomes) to establish consistency of measurement (i.e., harmonization; for a model-based empirical demonstration, see Drislane & Patrick 2017). The availability of these alternative scale sets provides a means to extend what we know about the neurophysiological, task-behavioral, and clinical-symptom correlates of key biobehavioral constructs, and to establish formal multimodal measurement models for these constructs, through use of existing large-scale data sets—including etiologically informative (e.g., twin, longitudinal) data sets (Patrick et al. 2019a).

A second point is that self-report scale operationalizations of the biobehavioral traits depicted in **Figure 3** represent distinct configurations of lower-order traits from conventional personality models. For example, scale-assessed threat sensitivity (boldness) encompasses specific facet traits from FFM domains of extraversion (E) and neuroticism (N) (negative) and, to a lesser extent, agreeableness (A) (negative) (Drislane et al. 2018; see also Poy et al. 2014). The reason is that the biobehavioral trait of threat sensitivity manifests itself in different psychological realms: social behavior (E, A), affective experience (N), and activity preference (E) (Kramer et al. 2012, Patrick et al. 2019b). Evidence for this comes from research showing that scales assessing fear/fearlessness in these different realms relate in common to physiological defensive reactivity as a function of the variance they share (Kramer et al. 2012, Vaidyanathan et al. 2009). Other work has shown that the trait of inhibitory control (disinhibition) indexes biological liability for externalizing problems more effectively when assessed in triarchic terms as opposed to FFM (as NEO Conscientiousness; Joyner et al. 2020) or AMPD terms (as PID-5 Disinhibition; Bowyer et al. 2020). These lines of evidence suggest that self-report-assessed traits need to be structured in distinctive ways to connect effectively with biological variables.

A further point is that self-report indicators of biobehavioral traits are intended to serve only as provisional referents for identifying candidate indicators from other modalities (e.g., lab-task performance, brain physiology) in establishing multimodal measurement models (Patrick et al. 2019a). Self-report trait scales are convenient and effective as referents because they can be constructed using different item sets to be indicative of specific psychological content, psychometrically reliable, and harmonized in terms of their convergence with one another and their relations with external criterion measures. Brislin et al. (2019) illustrated this by developing a scale measure of disinhibition from items available in the European IMAGEN data set (Schumann et al. 2010) and, using data for a different sample, showing that it correlated highly with a previously validated disinhibition scale and comparably well with an established neural indicator of disinhibition (oddball P3 response). The view of self-report scales as provisional indicators connects in turn to the idea of biobehavioral traits as open constructs that can be refined or reshaped based on empirical findings. As depicted in **Supplemental Figure 2**, Venables et al. (2018) illustrated this idea by specifying a structural model in which measures from three modalities (self-report, task performance, brain response) loaded onto lower-order modality factors, which loaded onto a common higher-order factor reflecting multimodal inhibitory control. A shift in the construct being measured—away from the modality of self-report—was evidenced by a smaller loading of the self-report modality factor onto the higher-order factor compared to loadings for the behavioral and brain modality factors.

A final key point regarding the framework depicted in **Figure 3** is that it includes a developmental component, depicted by the downward-pointing arrow on the right side. The literature on temperament indicates that traits are moderately stable across time but are affected by maturation and ongoing environmental events that alter psychobiological function (Bornstein et al. 2015, Kopala-Sibley et al. 2018). The types of measures that can be used to index traits also vary

across time. For example, self-report measures cannot be used with very young children, and tasks suitable for assessing behavioral and brain response often differ across age levels. Also, physiological or behavioral measures that relate concurrently to trait levels or clinical symptoms may not exhibit relations prospectively if they reflect dysfunctional states or persisting consequences of dysfunction, as opposed to dispositional liability (Perkins et al. 2020). I believe these complexities in characterizing the nature of biobehavioral risk for mental health problems, and pathways from risk to symptom expression in differing forms, can ultimately be addressed through systematic collaborative efforts along coordinated lines. Researchers dedicated to the scientific study of psychopathy are encouraged to benefit from and contribute to this broader mission in ways highlighted within this review.

SUMMARY POINTS

1. Psychopathy is a continuous dimensional and multifaceted condition marked by affective-interpersonal and impulsive-externalizing symptoms.
2. The triarchic model views the symptoms of psychopathy as expressions of three distinct trait dispositions: boldness, meanness, and disinhibition. These traits are conceptualized in biobehavioral-developmental terms, and their nomological networks include variables from behavioral and neurophysiological as well as self-report and clinical-diagnostic modalities.
3. Cleckley's (1976) characterization of psychopathy in psychiatric inpatients (as a "masked" form of insanity) emphasized features associated with boldness and disinhibition, whereas descriptions of criminally psychopathic individuals (as violent predators) emphasized features associated with meanness and disinhibition.
4. Based on cluster analyses of personality and clinical-symptom measures, variants (subtypes) of psychopathy have been identified in adult prisoner and nonprisoner samples and more recently in younger samples—distinguished in particular by low versus high levels of negative affect (e.g., anxiousness, mistrust, hostility). The new trait-dimensional system for diagnosing personality disorders in the fifth edition of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5) includes a specifier for antisocial personality disorder that reflects features of the low-anxiety (primary/bold) type.
5. Relations of symptom facets of psychopathy with other mental disorders can be understood in triarchic model terms. Boldness, which relates most to the interpersonal facet of psychopathy, is associated with a lack of internalizing symptoms and somewhat enhanced proneness to externalizing problems, whereas meanness (related most to psychopathy's affective facet) is associated with antagonistic externalizing symptoms in particular. Disinhibition, which relates most to psychopathy's impulsive-irresponsible facet, is associated with elevations in both disinhibitory externalizing and internalizing (particularly distress) symptomatology.
6. Research focusing on the triarchic model traits has also helped to clarify diverging relations of psychopathy symptom facets with biological and task-behavioral measures. Boldness, theorized to reflect low threat sensitivity, accounts best for psychopathy's relations with deficient fear reactivity, whereas disinhibition accounts best for its relations with impairments in cognitive-task performance and brain response. Meanness shows unique associations with deficits in fear-face recognition and reactivity.

FUTURE ISSUES

1. Further research is needed to examine the clinical and biobehavioral characteristics of individuals with extreme scores on different combinations of psychopathy-related traits as specified by the triarchic model or the DSM-5 dimensional-trait system.
2. Another important goal for future research is to refine and coordinate dimensional assessments of aggressive and nonaggressive antisocial deviance in youth to permit better integration of findings within and across age levels.
3. Research on psychopathy in youth has only recently begun to examine how the triarchic traits of disinhibition and boldness contribute to the emergence and expression of conduct problems, relative to and in concert with callousness-unemotionality (meanness). More research of this kind is needed.
4. Studies focusing on neurobiological systems and processes in psychopathy need to devote greater attention to developmental shifts in relations among biology, traits, and symptoms and to maturational and environmental factors that contribute to these shifts.
5. Drawing on extensive efforts that have been made to develop harmonized scale measures of the triarchic model traits using items from different inventories, research on psychopathy stands to benefit greatly from already-existing, large-scale specialized data sets—including longitudinal and/or twin data sets that contain variables from multiple measurement modalities.

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